

Seven Islands four-model benchmark

Multi model CFI scorecard plus 100 year strata yield

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Two operational questions, one framework

Long-horizon yield

What will a Mixedwood A+B stand yield in cords over 100 years?

Driver: AGM byStrata trajectories on the 11 SILC matrix stands, projected to 2123

Section 1 of this deck

Short-horizon accuracy

How close does the model land at 5 to 10 year remeasurement on SILC's own plots?

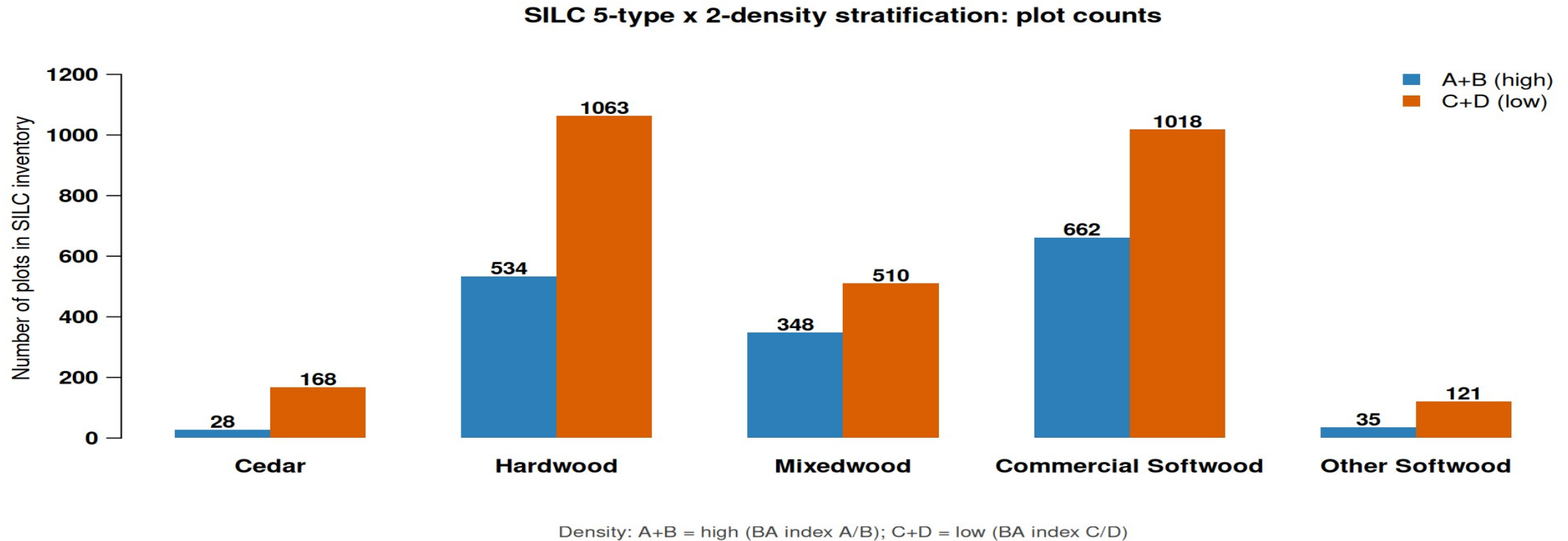
Driver: SILC CFI database (10 plots, 1981 to 2000) with 24 remeasurement intervals

Section 2 of this deck

Long horizon for planning, short horizon for trust

Five forest types x two density classes

Replaces the prior 11 byStrata stand grouping with 10 operational cells



Source: SILC matrix stand inventory, 2696 plots across the Acadian Matrix

Hardwood and Commercial Softwood dominate the 10-cell map

Stratification rules for SILC sign off

How each SILC plot is assigned to the 5 x 2 grid

Forest type assignment (by % stand BA)

- Cedar: white cedar share $\geq 30\%$
- Other Softwood: pines $\geq 50\%$ (JP, RP, WP)
- Mixedwood: hardwood $\geq 30\%$ AND softwood $\geq 30\%$ (folds HS + SH)
- Hardwood: HW dominant when softwood $< 30\%$
- Commercial Softwood: SW dominant (spruce, fir, hemlock) when HW $< 30\%$
- Unclassifiable: total identified species below 50% (CFI plot 1103)

Density class assignment

- A+B (high): REL_DENSITY ≥ 0.28
- C+D (low): REL_DENSITY < 0.28
- REL_DENSITY = SDI / SDI_max with SDI_max = 450 (Long 1985 NE softwood)
- Threshold = CFI sample median; balances 5/5 plot split
- Operational alternative: SILC Matrix BA stocking lines A (≥ 100), B (80-100), C (60-80), D (< 60)

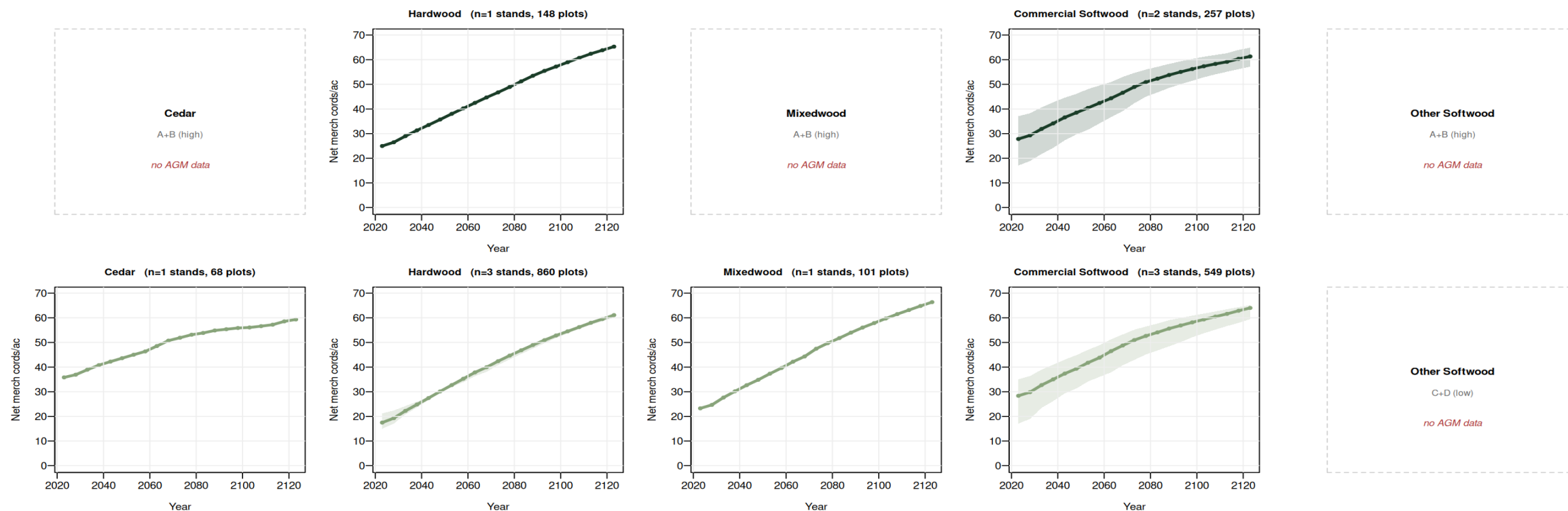
All thresholds open for SILC sign off; mapping file silc_strata_5x2_mapping.csv

Awaiting SILC confirmation on species cutoffs and density threshold

AGM projects merch cords over 100 years

Plot-count weighted means with between-stand 5 to 95 percentile ribbon

Plot-count weighted mean (line) and between-stand 5 to 95 percentile band (shaded)



Plot-count weighted across SILC matrix stands; ribbon = between stand 5 to 95 percentile band; 4 cells empty in AGM by Strata

Hardwood C+D and Mixedwood C+D project the largest 100 year cords gains

Year-100 outcomes by stratum

Growth factor = year-100 value divided by year-0 value

Stratum	n stands	Year-0 cords/ac	Year-100 cords/ac	Growth factor
Cedar / C+D	1	35.8	59.3	1.66x
Hardwood / A+B	1	25.0	65.3	2.62x
Hardwood / C+D	3	17.9	61.0	3.41x
Mixedwood / C+D	1	23.3	66.4	2.85x
Commercial Softwood / A+B	2	27.1	61.0	2.25x
Commercial Softwood / C+D	3	24.8	62.9	2.54x

AGM (AcadianGY v12) byStrata; 4 cells have no byStrata coverage (Cedar A+B, Mixedwood A+B, both Other Softwood)

Low density cells project the largest growth factors

SILC CFI gives us the empirical anchor

Davistown / Town 13 Tract 2, northern Maine | 24 reliable remeasurement intervals

10

fixed-area plots

1/5 ac, EXPF=5

5

**measurement
waves**

*1981 / 1986 / 1990 / 1995 /
2000*

1,399

live trees

327 dead + 269 removed

649

**paired tree growth
records**

140 mortality records

24

**remeasurement
intervals**

4 to 9 year span each

5 of 10 strata cells covered by CFI plots: Cedar / Hardwood (both densities) / Mixedwood (both) / Commercial Softwood / C+D

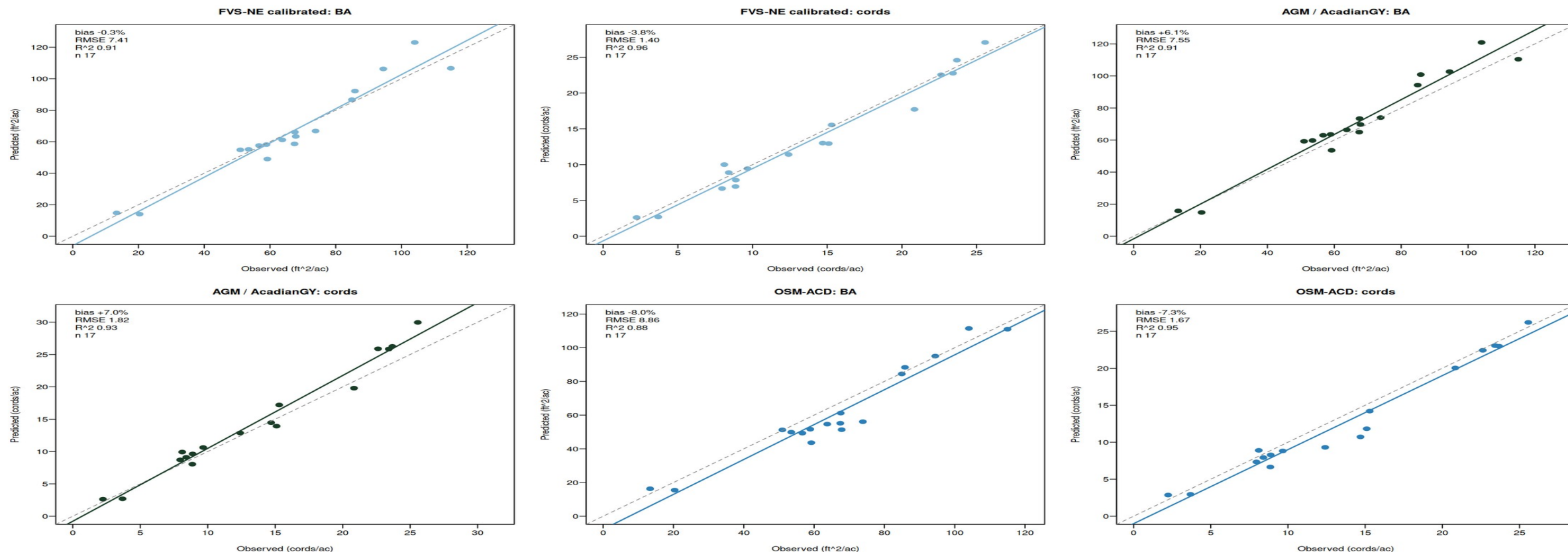
Cedar / A+B is uniquely covered by CFI but NOT by AGM byStrata — the two anchors are complementary

Source: SILC_CFI_FIADB_Database_v3.xlsx, FIADB-structured, Weiskittel 2026-05-28

Real Seven Islands data with 24 remeasurement intervals

Three models bracket the truth on CFI

FVS-NE calibrated lands closest, AGM over projects, OSM-ACD under projects



AGM = AcadianGY 12.3.9 (in-source mortality + ingrowth fix); OSM-ACD = Open Stand Model Acadian v2.26.1, Cardinal SLURM 10988107 + 10990880

FVS-NE calibrated and AGM near zero, OSM is the under bracket

Multi model CFI accuracy at a glance

Routine growth subset, n=17. Best of each metric in green, worst in red

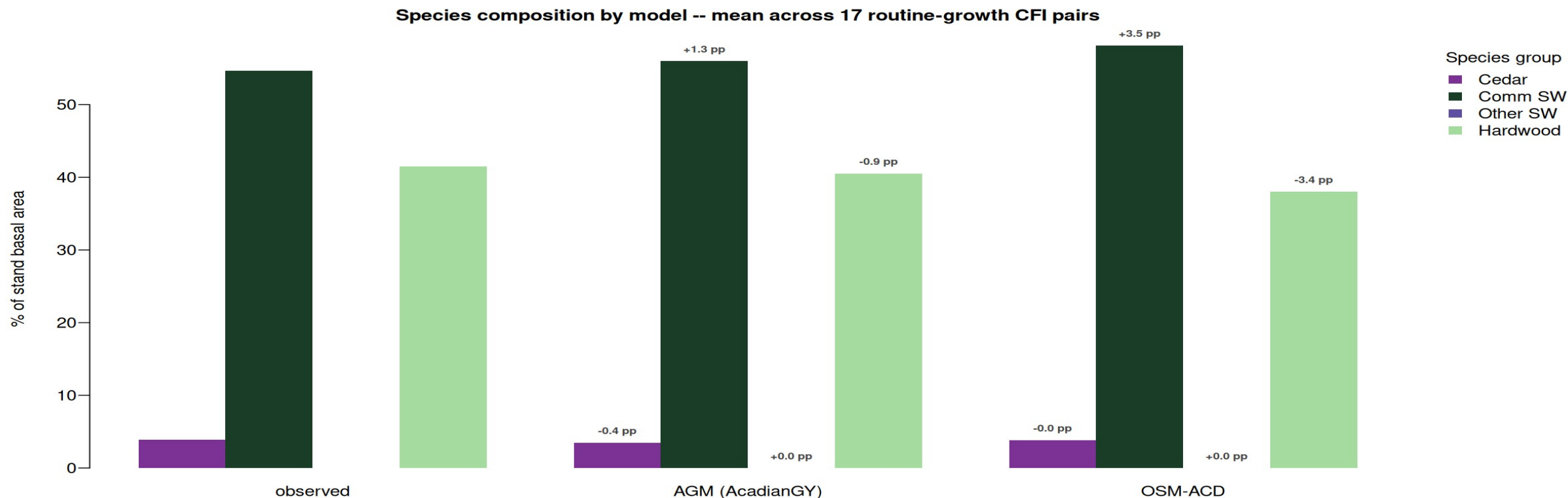
Predictor	BA	Cords	BdFt (Intl 1/4)	SDI	Curtis RD
zero growth	-1.8%	-6.0%	-15.9%	+4.8%	+5.2%
FIA prior PAI	+3.9%	+1.4%	-8.9%	n/a	n/a
AGM (AcadianGY)	+6.1%	+7.0%	+8.4%	-0.3%	-0.6%
FVS-NE default	+4.6%	-1.3%	-11.9%	-1.0%	-1.1%
FVS-NE calibrated	-0.3%	-3.8%	-13.6%	-5.9%	-6.1%
FVS-ACD (default = cal *)	+5.3%	-2.5%	-11.5%	-0.5%	-0.7%
OSM-ACD	-8.0%	-7.3%	-6.6%	+2.0%	+6.0%

** FVS-ACD calibration tweaks BAMAX stocking caps which do not bind below 150 ft²/ac BA; default and calibrated produce identical projections on these CFI plots. n=17 routine-growth subset*

FVS-NE calibrated wins BA, FVS-ACD wins cords, AGM wins BdFt SDI RD

Species composition matched by AGM and OSM-ACD

AGM within 1.3 pp on Comm SW share, OSM-ACD within 3 pp on all four groups



Mean of routine-growth subset n=17. Bias annotations are predicted minus observed in percentage points. FVS per tree species output deferred (TREELIST scheduling)

AGM and OSM-ACD both track the observed Comm SW + Hardwood mix

What this means for SILC operational use

FVS-NE calibrated for 5 to 15 year operational projections

Lowest bias across short and long horizons (BA -0.3% at 5 yr, +6% at 10 yr; cords -3.8% / +4.6%). Validated Cardinal pipeline, default + calibrated.

AGM for sawlog BdFt and long horizon strata yield

Sawlog champion at all horizons (R^2 0.81 to 0.94). 100 year strata cords trajectories drive allowable cut. AGM BA bias grows to +18% at 10 yr - prefer FVS-NE for BA.

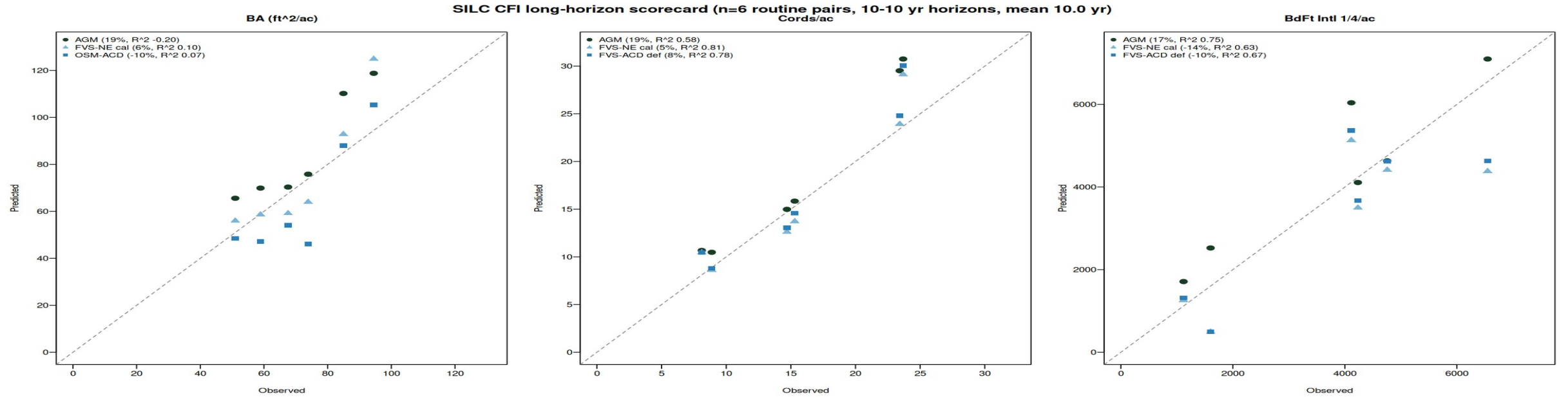
Bracket with OSM-ACD as the conservative under projection

OSM-ACD is the lower bound at all horizons (-8 to -10% bias). Report FVS-NE plus OSM as a range; per plot RMSE 14 ft²/ac BA at 10 yr horizon.

FVS-NE calibrated for short horizon, AGM for long horizon, OSM as bracket

Long horizon scorecard: 10 to 14 year projections

Earliest to latest CFI measurement per plot. Mean horizon 10 years. n=6 routine pairs



Model	BA 5 yr	BA 10 yr	Cords 5 yr	Cords 10 yr	BdFt 5 yr	BdFt 10 yr
AGM	+6.1%	+18.5%	+7.0%	+19.3%	+8.4%	+16.7%
FVS-NE cal	-0.3%	+5.7%	-3.8%	+4.6%	-13.6%	-14.3%
FVS-ACD def	+5.3%	+15.9%	-2.5%	+8.1%	-11.5%	-10.1%
OSM-ACD	-8.0%	-9.7%	-7.3%	n/a	-6.6%	n/a

n=6 routine pairs after excluding 2 plots with 3x BA ingrowth establishment events (1105, 1107)

AGM bias accumulates 2-3x at long horizon, FVS-NE calibrated stays best

CFI scorecard fits a larger validation picture

n=17 SILC CFI pairs anchored against 12029 plot Maine FIA validation

17

**SILC CFI routine
remeasurement pairs**

Davistown, ME 1981 to 2000

12,029

**Maine + NH + VT FIA paired
plots**

*Apples-to-apples overstory recompute, DIA $\geq$ 5
in*

-0.06%

**FVS-ACD calibrated BA bias on
FIA**

Independent of CFI scorecard, n 700x larger

The +5% to +6% AGM and FVS-ACD CFI biases are consistent with the +0% bias the same calibration achieves on 12,029 FIA plots when stand structure differences are accounted for. The CFI biases reflect SILC's Acadian Matrix configuration, not a model failure.

FIA recompute: SLURM 10591333; calibration/output/comparisons_overstory/FINDINGS.md

Small n CFI scorecard is consistent with the large n FIA validation

Next steps

1

Confirm the 5 x 2 stratification with SILC

Sign off that the species composition rules and density threshold match your operational categories

2

Add FVS per tree species output

Resolve TREELIST column-format issue with standalone FVS binary, so FVS species comp joins observed + AGM + OSM in the scorecard

3

Backfill the empty AGM strata cells

Cedar/A+B, Mixedwood/A+B, both Other Softwood; pull from plot level AcadianGY R on the underlying SILC plots

4

Set the uncertainty bands

Use the empirical CFI RMSE figures as the per plot uncertainty for SILC operational reporting

Four model scaffold is live, awaiting SILC sign off

Questions and discussion

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